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The interplay of engagement and learning regulation in online learning

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Abstract. This study is of significant importance as it examines the interplay between student engagement and self-regulated learning (SRL) in online learning environments. We analyse longitudinal data comprising student survey responses and log files from a learning management system (LMS). The participants were 165 students enrolled in a Finnish higher education institution course. Data collection included trace data from the LMS, detailing each student's actions with timestamps, and survey data collected at multiple points throughout the course. Students were classified based on their engagement patterns and self-regulation abilities. We utilised clustering methods to group students into engagement and SRL clusters. The findings, which are of great relevance, indicate a positive association between student engagement and SRL, suggesting that enhancing SRL skills can lead to improved student engagement and academic performance. The study, therefore, underscores the need to develop effective learning strategies and interventions that support students' self-regulation and engagement in online courses.

Keywords: Learning analytics, Process mining, Engagement, Self-regulated learning.

1 Introduction

This study addresses the challenge of understanding the interplay of student learning by presenting an approach to analysing longitudinal data on student engagement and self-regulation. We use survey data and log files to classify students based on their engagement patterns and self-regulation abilities. We aim to uncover new insights into the dynamics of student learning, which can inform the development of more effective learning strategies and interventions. This study uses longitudinal data to explore how learning evolves during a course.

2 Background

Engagement is a construct integrating three dimensions: behaviour, emotion, and cognition [2, 3]. Behavioural engagement includes, e.g., study efforts such as attendance, participation, and commitment to schoolwork. Emotional engagement explains a learner's positive and negative feelings about school and other students. Cognitive engagement characterises students' efforts to get a grip on learning tasks and concepts and the self-regulation of their learning process [4]. For this study, the behavioural dimension is the most important because it can be traced with the help of data gathered from a learning management system (LMS) using variables such as assignments completed, login frequency, number and frequency of postings, time spent creating posts, time spent online, number of learning resources accessed [5, 6]. [1, 7] have given an example of the methodology that can be used to measure online engagement.

The previous studies about the evolution of engagement have been two-fold. Studies show that previous engagement can predict future engagement [8, 9], but engagement is not constantly increasing [10]. Numerous factors, e.g., locus of control, self-esteem, parental expectation, communication, and the student's socio-economic status and grade history [11]. Furthermore, [12] has found that emotions, personality characteristics, prior learning experiences, shared values across contexts, and extra-curricular activities affect engagement.

When LMS data can be used to track the behavioural aspect of learning, it does not give the whole picture of what is happening during the learning process. Therefore, we need an understanding of self-regulated learning (SRL). SRL is often described as a process that includes setting goals, monitoring progress, and regulation of learning [13, 14]. Different SRL models focus on different parts of learning, such as motivation [15], metacognitive aspects [16], effective cognitive strategies [17], and how the regulation can be shared [18]. Some models are also considered general, as they value motivation and metacognition as essential elements of learning [14, 19]. There are different ways to measure this critical element of learning. One way to measure SRL is with the help of surveys. There are various surveys [20–24] which differ in the aspects of learning they focus on. Thus, these instruments can be used to study how students perceive their current state of learning.

Self-reported SRL data can be combined with trace data captured from LMS to, e.g., predict academic performance [25]. The students who have good SRL skills have higher engagement [26]. They achieve better grades and can manage their study surroundings effectively [27]. The results of the previous studies show that the courses helping students improve their SRL will improve their engagement [28] and grades [29].

In previous studies, the longitudinal approach has been used to study engagement in relation to academic achievement [1]. The relation of engagement with SRL remains an active area of inquiry. Thus, by combining the self-reported survey data with the trace data captured from LMS, this study aims to provide a more holistic view of the learning process by examining the following research questions:

RQ1: What are the engagement and learning regulation states during an online course?

RQ2: What is the association between engagement and learning regulation states during an online course?
RQ3: How do the combined states evolve during an online course?

3 Methods

The data for this study were gathered from two sources: a) trace data captured from a course organised in Moodle LMS and b) longitudinal survey data from the same course. The course was organised at the Finnish higher education institution (HEI). The students were Finnish first year business students. The students in the group are mostly ca. 20 years old, although the exact age of the participants was not collected. The parallel datasets represent two perspectives on learning. LMS data illustrates the actions students (n=165) take while studying online, while survey data shows the students' perceptions of their learning process. LMS data includes ID, timestamp, and the context of every student's action during the course. The survey data was captured multiple times during the course. Each student responded to the survey at the beginning of the course and before every assignment submission. The HEI has approved the research, and each student has given informed consent for their participation.

Various validated SRL surveys [20–24] have numerous items. This is a good approach when developing and evaluating the validity of the inventory. However, the survey length becomes a burden for learners [30] if the survey is used to gather repeated measures during the course. Therefore, it is essential to keep the surveys short when performing longitudinal studies [31], especially when we want to track the development of competencies. Single-item surveys may be a feasible solution for this kind of research setting [32]. It is known that people don't report the development precisely retrospectively [33]. Therefore, the reporting should be done in real time.

The survey included 11 statements about different SRL-related factors. The statements were taken from previously validated instruments [21, 23, 24] and were selected to match the features, the pedagogical approach, and the logistics of the course context. The same approach has been previously used by Saqr & López-Pernas [34], who conducted a longitudinal study with repeated measurement design, where with a small sample size (n=18), they captured 689 observations. The small sample size can be adequate when focusing on the within-person variance [35–37]. This raises the question about the validity of such a survey, but it is found that single-item construct questionnaires are as 'reliable as their multi-item counterparts' [32].

The survey items are used as statement variables in the analysis and are called AskHelp, Criteria, Deadlines, Prethink, Schedule, SelfAssess, Standards, Strategies, Summary, Time, and Location. We performed a correlation test to check if there were correlations between survey variables. Time and Location variables had a very high correlation. Thus, we combined these variables by calculating a mean for these two variables and named the newly created variable Undistracted. This new variable was then used to replace variables Time and Location in the survey data set. These variables represent the student's perception of learning regulation.

Next, we will explain the analysis methods to answer the research questions of this study.

RQ1: What are the engagement and learning regulation states during an online course?

Both data sets were normalised to ensure comparability across different scales using the BBmisc package [38] to remove differences across course iterations. This will set the mean value for each variable to zero, making comparing different variables using different scales easier. We then used the normalised data sets to detect latent states for both sets. The LMS data was used to detect an engagement state, and the survey data was used to detect the learning regulation state. For this purpose, we used the MoEClust package [39] with the Expectation-Maximization (EM) algorithm to determine the optimal number of clusters. Various models were compared using the Bayesian Information Criterion (BIC). The VVV model with 3 clusters was selected based on BIC. The clusters were visualised using the prcr package [40].

Survey responses were similarly clustered. We suppressed different SRL variables to indicate the learner's state at different time points during the course [7]. The MoEClust package was again used to identify optimal clusters following a procedure similar to the one explained in the previous paragraph for the LMS data. This time, the VVV model with 2 clusters was selected based on BIC.

From here on, the two sets of clusters are referred to as states. The LMS clusters are referred to as course engagement states with levels Active, Average, and Disengaged, while survey clusters are referred to as learning regulation states with levels LowReg representing low regulation of learning and HighReg representing high regulation of learning.

RQ2: What is the association between engagement and learning regulation states during an online course?

To answer the second research question, we used the vcd package [41] to plot the relationship between categorical variables engagement and learning regulation states. We used the mosaic function to create a mosaic plot, a graphical representation of data from contingency tables. The results show the Pearson residuals for different relations along with p-values, which are used to evaluate the statistical significance of the results.

RQ3: How does the combined state evolve during an online course?

After the states were determined, we created sequences to represent the evolution of learning during the course. This was done by dividing the course into 11 segments. The segmentation was done with the help of survey submission timestamps. We wanted to combine the two states to represent the parallel learning development on survey data and LMS. Thus, we used the timestamps of the survey submissions to cut the LMS data into segments, representing the learning development within the course. We used an interval between two survey submissions for each student from their trace data to calculate the LMS variables for each student for each segment individually. This enabled us to track the process on an asynchronous course where explicit deadlines were not feasible solutions to divide the course into segments. This approach allowed the segmentation of the course data following the individual progression for each student, making it possible to do the between-student comparison. We used a graphical Gaussian models (GGM) variation of the multilevel vector autoregressive model (VAR) with the mlVAR [42] to plot the process map showing how students progress between different combined states during the course.

4 Results

The clustering analysis revealed distinct profiles of student engagement and self-regulation. The sequence analysis provided insights into the transitions between different states.

RQ1: What are the engagement and learning regulation states during an online course?

The engagement and learning regulation states are displayed in Figure 1. There are three states for the engagement. The Average state represents learners with average engagement values; their values are close to the mean value of normalised variable values. The disengaged state had below-average values on the number of active days, interactions with assignment-related activities, learning resources, and surveys. Number of sessions is very low, while the mean session length is very high. The active state represents learners with above an average number of active days, interactions with assignment-related activities, learning resources, and survey activities. These students visit LMS often, but the length of the learning sessions is below the mean.

There are two states for learning regulation. LowReg represents learners whose survey responses are below the normalised mean on every scale, and HighReg represents learners whose survey responses are above the normalised mean on each scale.

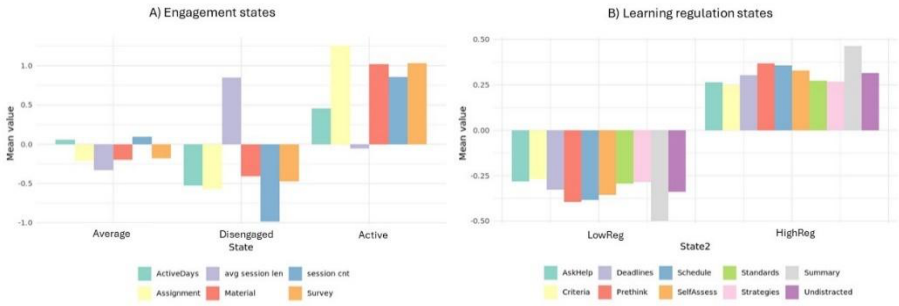


Fig. 1. Mean value of each variable for each cluster: A) engagement state, and B) learning regulation state.

RQ2: What is the association between engagement and learning regulation states during an online course?

The association between the Active engagement state and the High learning regulation state was positive and statistically significant. The blue colour in Figure 2 indicates this. Another positive and statistically significant association was found between the Disengaged engagement and Low learning regulation states. The negative and statistically significant associations (red colour) were found between a) Disengaged engagement state and High learning regulation state, and b) Active engagement state and Low learning regulation state. Thus, it can be summarised that high engagement is related to high learning regulation, and vice versa; low engagement is related to low

learning regulation. The table on the right of Figure 2 presents the number of students in different states at the beginning and at the end of the course.



State	Beginning	End
Active/HighReg	1 (0.6%)	46 (27.9%)
Active/LowReg	4 (2.4%)	21 (12.7%)
Average/HighReg	17 (10.3%)	36 (21.8%)
Average/LowReg	34 (20.6%)	41 (24.8%)
Disengaged/HighReg	31 (18.8%)	4 (2.4%)
Disengaged/LowReg	78 (47.3%)	17 (10.3%)

Fig. 2. Left: Mosaic plot showing the association between engagement states and learning regulation states during the course, right: the number of students in different states at the beginning and at the end of the course.

RQ3: How does the combined state evolve during an online course?

The process map (Figure 3) shows the transitions between combined engagement and learning regulation states. On average, it is more likely to remain in state transit from one state to another. Average LowReg was the most stable category, more likely to transition to the same state (68%) and least likely to change. If a student moves to another combined state, the most likely transition is to Average HighReg (14%). Average HighReg is the second most stable (62%) combined state, and the most likely transition from there is to Active HighReg (22%). Active HighReg is the third most stable (59%) combined state, and from there, the most likely transition is to Average HighReg (24%). We can conclude that the students with an average or active engagement are more stable to remain in their current state.

If the transition takes place, the most likely (42%) transition between combined states is from a Disengaged HighReg learning state to an Average engagement HighReg learning state. The self-loops for Average engagement LowReg learning (68%), Average engagement HighReg learning (62%), Active engagement HighReg learning (59%), and Active engagement LowReg learning (47%) have higher values than this most likely transition between states. Therefore, it seems more likely to stick with the previous state than move from one state to another. Notably, there is no direct link between Disengaged LowReg and Active HighReg combined states. It is extremely unlikely for a student to start Disengaged and with LowReg learning and then continue directly to Active engagement HighReg learning. If the transition between these two states happens during a course, it only occurs gradually between other mapped states.

The outer circle of the node shows the initial probability of starting the course from the particular combined state. Thus, the Disengaged LowReg learning state (47%) is the most common initial state, with around half of the circle filled. This is followed by Average LowReg (21%) and Disengaged HighReg (19%) states. Starting the course with an Active HighReg state (1%) is uncommon. The number of students belonging to different states is presented on right side of Figure 2.

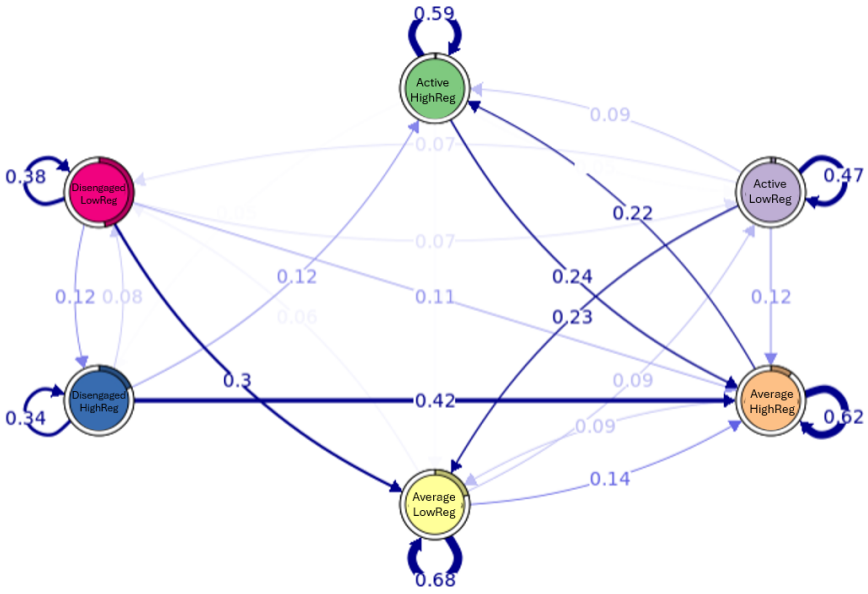


Fig. 3. Process map of dynamics between different states. Each node is a combined state of engagement and learning regulation. Edges represent the transition between two combined states, and the number indicates the likelihood (percentage) of the transition between the two combined states. The outer circle represents the initial probability of starting the course from that combined state.

5 Discussion and conclusion

The findings of this study are similar to the previous studies that have found a positive relationship between engagement and SRL [22]. The positive relation could be fostered by educating students on SRL. This has resulted in improved engagement [23] and academic achievement [20, 24]. SRL skills are vital for successful studies at the early stage of their studies at university [38]. Therefore, the developmental trajectory of the skills and how they could be enhanced should be studied. This would help to investigate whether the development could be fostered during the courses of the first semester of studies.

Our findings reveal novel, significant insights into the transitions between combined engagement and learning regulation states. Notably, the stability of low regulation states (Average LowReg at 68%, Active LowReg at 47%, and Disengaged LowReg at 38%) is a key observation. These likelihoods are higher than the most likely transition to different combined states. This stability suggests that students are more likely to remain in their current state than transition to another. On the other hand, when we assess the number of different states in the beginning and the end of the course (Figure 2), we can see that the number of students in disengaged states has lowered greatly. This suggests that the transition between states happens slowly during the course.

The implications of these findings are manifold. Firstly, understanding the stability of low-regulation states can inform the design of educational interventions [43]. For instance, targeted support could be provided to students in low-regulation states to facilitate their transition to higher-regulation states. This could include personalised learning resources or additional tutoring.

Our findings also underscore the need for sustained support and intervention strategies over time. The absence of a direct link between Disengaged LowReg and Active HighReg states suggests that transitions to more engaged and regulated states occur gradually. This highlights the importance of consistent support and interventions that are in line with monitoring the student's progress rather than a one-time solution.

Understanding the dynamics of state transitions is not just an academic exercise. It has a direct impact on student outcomes. By comprehending these dynamics, educators can identify students who may require additional support and intervene promptly [44].

Looking ahead, there are several avenues for future research. One promising direction is the further exploration of multichannel data [45]. We can gain a holistic understanding of engagement and SRL by combining data from various sources (e.g., LMSs, student surveys, etc.). If this data is analysed with a longitudinal approach [1], we can get a deeper understanding of the evolvement of states throughout a course. This could help in understanding how learning processes develop during a course and contribute to the broader field of learning analytics.

There are some limitations to this study. First, the data is gathered from a single course. The learning design might affect the student's actions on the course platform and their motivation, affecting the survey responses. Second, the students involved were a homogeneous group of Finnish students. To overcome these limitations, the study should be repeated in different contexts.

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